

DANYANG LIU

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RESEARCH INTERESTS

- Frontier VLMs and world models: architecture design, data-centric modeling, and evaluation.
- Multimodal agents for long-video editing, deep research, tool use, and product-facing AI workflows.
- Grounded and controllable vision-language generation from image sequences.

WORK EXPERIENCE

Reka AI, *Member of Technical Staff (Research Scientist)* Mar. 2025 – Present

- Trained production-grade vision-language models, contributing to architecture, data pipelines, training, and evaluation.
- Built multimodal agent products for long-video editing and deep research, spanning planning, tool use, video/audio understanding, retrieval, and deployment.

SELECTED INDUSTRY PROJECTS

Reka Edge: Efficient Multimodal VLM (app.reka.ai, [Hugging Face](https://huggingface.com)) Oct. 2025 - Feb. 2026

- Contributed to a 7B image/video-language model optimized for image understanding, video analysis, object detection, and agentic tool use.
- Worked on model architecture, training, evaluation, and multimodal data pipeline development.

World-Language-Action Model Mar. 2026 - Present

- Contributed to a dual-decoder world model that takes text, image, and video inputs and generates text, video, or action outputs for robotics.
- Worked on model design, data collection, data platform design, and multimodal training pipelines.

Reka Clip: Long-Video Editing Agents (creator.reka.ai) Jun. 2025 - Sep. 2025

- Built agents that turn hour-long videos into prompt-guided summaries and edited clips.
- Designed long-video segmentation, visual reframing, ASR integration, hierarchical chaptering, and planner-driven clip selection workflows.

Reka Research: Deep Research Agents (app.reka.ai) Mar. 2025 - May. 2025

- Built REACT-style agents for deep research over web sources and private documents, with multi-step search, self-reflection, document ingestion, and retrieval-augmented reasoning.

EDUCATION

University of Edinburgh, *Ph.D. in Natural Language Processing* Sep. 2020 - May. 2026

Thesis: *Character Grounding and Planning for Visual Story Generation*
Advisors: Frank Keller and Mirella Lapata

Shanghai Jiao Tong University, *M.Sc. in Information Security* Sep. 2017 - Mar. 2020

Southeast University, *B.Eng. in Information Engineering* Aug. 2013 - Jun. 2017

PUBLICATIONS

- [1] Eileen Wang, Hiba Arnaout, Dhita Putri Pratama, Shuo Yang, **Danyang Liu**, Jie Yang, Josiah Poon, Jeff Z. Pan, Soyeon Caren Han. [MMCOMET: A Large-Scale Multimodal Commonsense Knowledge Graph for Contextual Reasoning](#), submitted to the Conference on Neural Information Processing Systems (**NeurIPS**) 2026, **Under review**.
- [2] **Danyang Liu**, Mirella Lapata, Frank Keller. [Generating Visual Stories with Grounded and Coreferent Characters](#), Transactions of the Association for Computational Linguistics (**TACL**), 2025, Suzhou, China.
- [3] **Danyang Liu**, Fanjie Kong, Xiaohang Sun, Dhruva Patil, Avijit Vajpayee, Zhu Liu, Vimal Bhat, Najmeh Sadoughi. [Detect, Disambiguate, and Translate: On-Demand Visual Reasoning for Multimodal Machine Translation with Large Vision-Language Models](#), In Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (**NAACL**), 2025, Albuquerque, New Mexico.
- [4] Danna Zheng, **Danyang Liu**, Mirella Lapata, Jeff Z. Pan. [TrustScore: Reference-Free Evaluation of LLM Response Trustworthiness](#), Workshop on Secure and Trustworthy Large Language Models, (**SeT-LLM @ ICLR**), 2024, Vienna, Austria.
- [5] **Danyang Liu**, Mirella Lapata, Frank Keller. [Visual Storytelling with Question-Answer Plans](#), In Findings of the Conference on Empirical Methods in Natural Language Processing (**EMNLP Findings**) 2023, Singapore.
- [6] **Danyang Liu**, Frank Keller. [Detecting and Grounding Important Characters in Visual Stories](#), In Proceedings of the AAAI Conference on Artificial Intelligence (**AAAI**) 2023, Washington, DC.
- [7] Amr Keleg*, Matthias Lindemann*, **Danyang Liu***, Wanqiu Long*, Bonnie L Webber. [Automatically Discarding Straplins to Improve Data Quality for Abstractive News Summarization](#), In Proceedings of NLP Power! The First Workshop on Efficient Benchmarking in NLP (**ACL Workshop**) 2022, Dublin, Ireland.
- [8] **Danyang Liu**, Juntao Li, Meng-Hsuan Yu, Gongshen Liu, Rui Yan. [A Character-Centric Neural Model for Automated Story Generation](#), In Proceedings of AAAI Conference on Artificial Intelligence (**AAAI**) 2020, New York, USA.
- [9] Meng-Hsuan Yu, Juntao Li, **Danyang Liu**, Rui Yan. [Draft and Edit: Automatic Storytelling Through Multi-Pass Hierarchical Conditional Variational Autoencoder](#), In Proceedings of AAAI Conference on Artificial Intelligence (**AAAI**) 2020, New York, USA.
- [10] **Danyang Liu** and Gongshen Liu. [A Transformer-Based Variational Auto Encoder for Sentence Generation](#), In Proceedings of the International Joint Conference on Neural Networks (**IJCNN**) 2019, Budapest, Hungary.

INTERNSHIP EXPERIENCE

Amazon Prime Video, Seattle, U.S.

Jul. 2024 - Oct. 2024

Research Intern

- Research on Using Large Vision-Language Model for Multimodal Machine Translation.

Huawei Edinburgh Research Centre, Edinburgh, U.K.

May. 2021 - Oct. 2021

Research Intern

- Research on Jointly Learning Conversational Semantic Parsing and Answerability Detection.

SELECTED AWARDS

School of Informatics CDT NLP PhD Scholarship <i>4-year scholarship from the University of Edinburgh</i>	2020 - 2024
First-Class Postgraduate Academic Scholarship <i>3-year scholarship from Shanghai Jiao Tong University</i>	2017 - 2020
Outstanding Graduate of Shanghai Jiao Tong University	2020
Meritorious Winner of The Interdisciplinary Contest in Modeling	2017
Excellent Graduation Project (5%) of Southeast University	2017